

Design Basis – Foundations and Electrical Balance of Plant

Harrow Bay Offshore Wind · Revision C (approved for construction)

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1. Introduction

This document defines the design basis for the foundations, support structures and electrical balance of plant of the Harrow Bay Offshore Wind farm. It is the governing reference for the detailed design carried out by Rhenus Steelworks Foundations (monopiles and transition pieces), Norfab Offshore Structures (HB-OSS) and Severn Subsea Cables (export and array cables), and for the integrated load analysis with Halden Turbine Systems.

2. Project overview

Parameter	Value
Installed capacity	480 MW
Turbines	40 × Halden HTS-12, 12 MW, rotor 236 m, hub height 140 m above MSL
Foundations	Monopile with bolted transition piece, 9.5 m diameter at mudline
Water depth	28.0 – 36.0 m LAT
Distance to shore	38 km
Array cables	66 kV, ~ 95 km, 7 circuits
Export	2 × 220 kV HVAC, 62 km each, to Harrow Bay East
Design life	30 years (foundations), 25 years (turbines, extendable)

3. Design standards

- DNV-ST-0126 Support structures for wind turbines (2021)
- DNV-ST-0437 Loads and site conditions for wind turbines
- IEC 61400-3-1 Design requirements for fixed offshore wind turbines
- DNV-RP-C203 Fatigue design of offshore steel structures
- DNV-RP-B401 Cathodic protection design
- EN 1993 (Eurocode 3) where referenced by DNV standards
- IEC 60840 / IEC 63026 for 220 kV and 66 kV submarine cables

4. Metocean conditions

Metocean parameters are taken from the site-specific metocean study (HB-ENG-REP-0011), based on 42 years of hindcast data calibrated with a floating LiDAR and wave buoy campaign from March 2023 to February 2024.

Parameter	1-year	50-year	Unit
Significant wave height Hs	6.1	9.8	m

Peak period T_p (associated)	10.4	13.5	s
Maximum wave height H_{max}	11.3	18.2	m
10-min mean wind speed at hub height	33.0	44.6	m/s
Depth-averaged current speed	0.9	1.1	m/s
Highest astronomical tide (HAT) above LAT	–	4.6	m
Storm surge (positive)	0.9	1.7	m

Mean annual wind speed at 140 m is 10.1 m/s with a Weibull shape factor of 2.2. Marine growth is assumed as 100 mm between +2 m and -10 m LAT and 50 mm below.

5. Geotechnical conditions

The site investigation comprised 40 seabed CPTs to 45 m, 12 boreholes and a geophysical survey of the entire array area. Soils consist of a thin veneer of loose to medium dense sand (Unit 1, 0–3 m) over dense to very dense sand (Unit 2) and stiff to very stiff glacial clay (Unit 3, Bolders Bank / Egmond Ground equivalent) from 8–15 m below seabed. Boulders are present in Unit 3; see the geotechnical interpretative summary.

6. Monopile and transition piece design

Item	Design value
Monopile diameter	9.5 m (constant), 7.0 m at top flange via cone
Wall thickness	75 – 115 mm
Length	71 – 82 m (position specific)
Embedded length	33 – 38 m
Mass	1,160 – 1,420 t
Steel grade	S355ML / S420ML to EN 10225
TP height / mass	26.5 m / 365 t
Connection	Bolted L-flange, 148 × M72 bolts
Verticality tolerance	$\leq 0.25^\circ$ after installation
Corrosion protection	Coating + sacrificial anodes (aluminium) on TP anode cage
Scour protection	Rock armour, pre-installed filter layer at strings B and C

The 160 dB SEL noise limit at 750 m (consent condition 16) drove the decision for a lower maximum hammer energy (3,000 kJ) and slightly longer piles at positions with dense sand (Unit 2).

7. Electrical design basis

- Offshore substation HB-OSS with 2 × 280 MVA 220/66 kV transformers, 66 kV GIS with 7 circuit breakers for array circuits
- Export cables: 3-core 220 kV XLPE, 1,000 mm² copper, fibre optic integrated, burial depth 1.0–2.0 m
- Array cables: 66 kV XLPE, 630 mm² (first sections) and 240 mm² aluminium, burial target 1.0 m
- Reactive compensation onshore at Harrow Bay East (STATCOM 2 × 60 Mvar)
- Grid compliance per NordGrid Transmission grid code and connection agreement

8. Interfaces

Interface	Party A	Party B	Document
Tower flange / TP top flange	Halden	Rhenus	HB-IF-001
J-tube / cable entry	Rhenus	Severn	HB-IF-004
Cable hang-off at OSS	Norfab	Severn	HB-IF-007
SCADA / fibre	Halden	Norfab	HB-IF-011
Grid connection point	Weyland	NordGrid Transmission	HB-IF-015